Temporal Trends in Psychometric Reporting Quality in Psychology

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Abstract

Sound measurement is fundamental to psychological science, yet the transparency with which psychometric instruments are developed, validated, and reported is unclear. Although the past two decades have seen the publication of formal reporting standards and a broad movement toward methodological reform following the replication crisis, whether these developments have improved the reporting of newly developed and modified measures remains unknown. The present study examined whether the quality of psychometric reporting changed across the period 2002 to 2026, using a large sample of 1,142 measures nested within 530 articles drawn from five psychology subfields. Reporting quality was operationalised as the proportion of preregistered coding criteria met for each measure, and modelled using a Bayesian multilevel beta regression with a piecewise linear effect of publication year, incorporating a breakpoint fixed at 2014 to capture any shift associated with the revised reporting standards and the reform period that followed. Reporting quality did not credibly change over the study window, either overall or within any of four sub-dimensions comprising scale construction, sample description, validity and reliability, and dimensionality, and no credible change occurred at the 2014 breakpoint. Reporting was uniformly low and differed markedly across sub-dimensions, from sample description at the highest level to validity and reliability evidence at the lowest, though this ordering remained stable throughout. Variation in reporting quality was attributable to differences between articles rather than to publication year. These findings indicate that methodological reform has not reached the reporting of measurement, and suggest that structural, enforced interventions are now required.

Temporal Trends in Psychometric Reporting Quality in Psychology

A fundamental aspect of psychological science is defining and measuring what is being studied. Before the aetiology of depression can be investigated or the efficacy of a treatment evaluated, the construct must be defined and instruments capable of measuring it accurately must be built (Flake & Fried, 2020). This is difficult because constructs such as depression, personality traits, attitudes, and cognitive abilities rarely permit direct observation. Construct validity, the degree to which a measure's scores track a person's true standing on the underlying construct, is therefore foundational to psychological measurement (Clark & Watson, 2019; Cronbach & Meehl, 1955). Because it can only be judged when authors report sufficient detail about their measures, incomplete reporting undermines both the assessment of construct validity and the interpretability of research claims (Lawson et al., 2026). When construct validity is poor, scores do not reflect the attribute they are claimed to, and the resulting inferences are compromised in a way that no amount of rigorous design, advanced statistics, or large samples can correct (Flake & Fried, 2020; Loevinger, 1957). The quality with which measurement is conducted, and the transparency with which it is reported, is thus not one methodological concern among many but a precondition for the accumulation of psychological knowledge.

The state of psychometric reporting

Despite measurement's foundational role in study validity, important information about it is routinely absent from published psychology articles. Measures are used without reference to their source, evidence that they capture the intended construct is often lacking, and unjustified measurement flexibility abounds. Barry et al. (2013) reported that 40% to 93% of measures across seven health education journals lacked validity evidence, and Weidman et al. (2017) found that 69% of 356 measurement instances in emotion research included no reference to prior

research or a systematic development process. Flake et al. (2017) similarly reported that 40% of measures gave no indication of their source, 19% omitted item count, and 9% omitted the response scale, with reliability, where reported, typically limited to Cronbach's alpha alone.

The problem deepens once reporting is judged not by what is disclosed but by what is true. Hussey and Hughes (2020) subjected 15 widely used self-report measures to comprehensive validity assessment, including factor structure, test-retest reliability, and measurement invariance. Only 4% demonstrated good validity, though 88% appeared adequate on internal consistency alone, a pattern the authors termed hidden invalidity: the one property psychologists report regularly is close to useless as a guide to a measure's validity. A subsequent review reinforced this, finding only about 9% of multi-item scales in the Reproducibility Project studies were accompanied by any factor analysis (Flake et al., 2022).

These concerns extend beyond self-report questionnaires. Parsons et al. (2019) argued that psychology lacks a tradition of reporting the reliability of cognitive and behavioural task measures, despite this being a precondition for robust inference. That the problem now has its own literature is telling: Meier (2023) framed an entire collection around the persistence of measurement problems. Together, this work shows inadequate psychometric reporting is broad and persistent across subfields, measure types, and evidence categories. What it does not establish is whether the situation has changed over time, since almost every study offers a cross-sectional snapshot rather than tracking whether reporting quality has improved, stagnated, or worsened.

Questionable measurement practices and researcher degrees of freedom

Flake and Fried (2020) coined "questionable measurement practices (QMP)" for decisions that raise doubts about a measure's validity, and ultimately about a study's conclusions.

Measurement, they argued, has its own “garden of forking paths”: a ten-item questionnaire can be summarised into a single sum score in over a thousand ways, with further options once subscales and alternative analyses are considered. This flexibility exists regardless of intent, and when the decisions taken go unreported and unjustified, the field cannot evaluate the study's internal, external, statistical-conclusion, or construct validity. QMP are in this sense broader in consequence than the questionable research practices that dominated reform, since they compromise the data themselves rather than just their analysis.

The Standards for Educational and Psychological Testing, revised in 2014 by the American Educational Research Association, the American Psychological Association, and the National Council on Measurement in Education (American Educational Research Association, et al., 2014), remain the most comprehensive framework for validity evidence, organised around test content, response processes, internal structure, relations to other variables, and testing consequences. The revised Journal Article Reporting Standards codified expectations for transparent measurement reporting (Appelbaum et al., 2018) , and more recent guidance has continued to systematise the reporting of quantitative analysis at both the planning and results stages (Schubert et al., 2025). The existence of such standards, however, guarantees nothing about their uptake. A guideline can be published without being read, cited without being followed, and endorsed in principle while ignored in practice. Whether the reporting behaviour of working psychologists shifted in the wake of these frameworks is a question that has gone largely unexamined.

Measurement and the credibility revolution

This question matters because measurement quality is directly linked to replicability. Poor measurement undermines replication through attenuation: when an instrument lacks

precision, a study's signal-to-noise ratio falls, statistical power declines, and the probability of failing to detect a genuine effect rises across studies (Loken & Gelman, 2017). Hedge et al. (2017) showed formally that measures with poor reliability produce highly variable effect-size estimates, making successful replication a near-impossibility even when the underlying effect is real. Inadequate reporting compounds this: a field that cannot observe its measures' reliability cannot diagnose measurement error as a cause of non-replication, since that is exactly the information most often withheld.

The replication crisis, marked by the Open Science Collaboration's (2015) finding that only 36% of 100 published psychological studies replicated, prompted sweeping methodological reform. Preregistration, registered reports, open data, open materials, and the Transparency and Openness Promotion guidelines spread across journals and subfields within a few years, a period Korbmacher et al. (2023) characterised not as a collapse but a credibility revolution of structural, procedural, and community-driven change. If this revolution reshaped the field's methodological culture, its influence might reasonably be expected to have reached measurement practice too.

The most direct evidence on this point is not encouraging. In the study most comparable to the present one, Lawson et al. (2026) coded 1,198 questionnaire measures across 506 *Social Psychological and Personality Science* articles published between 2010 and 2020 and examined whether measurement-reporting practices changed. Most did not. Reporting of scale names, citations, item counts, and reliability coefficients remained essentially flat, and the only practice to increase substantially was the availability of study materials. The authors concluded that increased attention to measurement reporting has not necessarily translated into better reporting. The reform movement, on this evidence, may have moved materials into the open without moving the reporting of measurement quality at all.

Two features of this finding define the space the present study occupies. First, it is powerful precisely because it is temporal, modelling change over time rather than offering another cross-sectional snapshot. Second, its scope was deliberately narrow: a single journal, subfield, and decade, coding only the lower-effort details authors can report, while noting that more demanding indicators like factor structure and convergent or discriminant validity are unlikely to appear when even these basics are missing. Whether that flat trajectory generalises across the wider field, a longer window, and the deeper dimensions of reporting left uncoded, remains open.

The present study

These literatures together define a clear gap. Evidence shows psychometric reporting is poor, standards exist prescribing what it should be, reform has plausibly had time to improve it, and the one study to test change directly suggests it has not. What is missing is a systematic, large-scale, multi-subfield examination of whether the quality of psychometric reporting for newly developed and modified measures has changed over time, and, if so, whether any change has been uniform across the dimensions of reporting or instead confined to particular kinds of evidence.

This gap matters more as measurement itself proliferates: ad hoc scales, online adaptations, and cross-cultural modifications have all grown alongside the field's output (Anvari et al., 2025). If poor reporting has persisted despite available guidelines, this threatens cumulative psychological science, since ever more theory rests on inadequately described measurement. If reporting has improved, this would show reform has reached measurement practice and could help identify which mechanisms worked.

The present study examines whether reporting quality for newly developed and modified measures changed between 2002 and 2026, spanning both the pre-reform era and the reform period following the replication crisis, using a large sample of articles across five subfields (clinical, applied, cognitive, developmental, social). It extends prior work by broadening scope beyond single-journal samples, lengthening the window to over two decades (from before the Standards' revision and credibility revolution to well after), and coding deeper reporting dimensions, including scale construction, participant description, validity and reliability evidence, and dimensionality, that earlier work acknowledged but did not examine.

The study addresses one primary and two secondary research questions. The primary question asks whether overall reporting quality improved across 2002 to 2026. The first secondary question examines the type of validity evidence reported (e.g., dimensionality, internal consistency, convergent and discriminant validity, relations with other variables), since improvement concentrated in some evidence types but not others would indicate whether the field is deepening its engagement with construct validity or merely meeting its least demanding expectations. The second asks whether development procedures (e.g., expert review, cognitive interviewing, piloting) have changed over time, addressing whether any shift reflects genuine change in how measures are built rather than only in how they are described.

Method

Description of the articles included

The final analytic sample comprised 530 articles reporting 1,142 psychometric measures, spanning a broad range of Journal Citation Reports (JCR) categories. The most represented were multidisciplinary psychology, education and educational research, and psychiatry, followed by the clinical, social, experimental, and developmental psychology categories, with a substantial

tail of others including health care sciences, criminology, and nursing. Because journals are often classified under more than one JCR category, article counts exceed the sample total; the full distribution appears in Table A1. Although the sampling frame was designed around five psychology subfields, the analysed sample is characterised here by journal category rather than the intended subfield strata.

Coders

Coding was carried out by a geographically distributed team of researchers based across multiple continents, including Europe, Australia, North America, South America, and Asia. Coders varied in career stage, ranging from master's and doctoral students to early- and mid-career researchers, with early-career researchers and postdoctoral researchers forming the largest part of the pool. Demographic information such as age and gender was not systematically recorded for individual coders and is therefore not reported here.

Materials

Data sources and eligibility criteria

Article metadata were drawn from OpenAlex, an open bibliographic database whose coverage exceeds that of Web of Science and Scopus by several orders of magnitude (Culbert et al., 2025), and which provides a documented, free application programming interface (API) supporting bulk extraction, filtering, and randomised sampling. Journal-level metrics were obtained from Clarivate's Journal Citation Reports (JCR), last updated on 15 October 2025 and indexing 5,969 journals. To avoid excluding interdisciplinary work published outside dedicated psychology outlets, journals were drawn from the JCR category groups Psychiatry/Psychology and Social Sciences, General.

Eligibility was assessed in two sequential stages. The first was embedded in the OpenAlex API call, operating on article-level metadata through structured queries, one per subfield, each identified by its Wikidata concept. To be retrieved, an article had to have been published between 2002 and 2026, possess a DOI together with an ISSN or eISSN, not have been retracted, be indexed within a subfield of psychology, be of document type "article", and be written in English. The second stage operated at the journal level: duplicate records were removed, and each article's ISSN or eISSN was cross-referenced against the compiled list of journals and their JCR metrics, with any article whose identifiers failed to match excluded. This ensured that every retained article could be linked to the journal-level covariates required for analysis.

Because the eligibility criteria identified well over one million papers, manual screening of all was infeasible. A randomised subset of 10,000 articles was therefore drawn for each of five subfields (clinical, applied, cognitive, developmental, and social), yielding a starting sample of 50,000 before the removal of duplicates and non-matching records. The minimum target for fully coded articles was approximately 1,000, conservatively mirroring comparable reporting-practice reviews (Barry et al., 2013). This paper reports the articles coded up to 8 August 2026; because the models fitted are Bayesian, no statistical issue arises from analysing the data before the planned sample size is reached.

Design

Each eligible article was coded against a structured scheme, and its coded responses were converted into quantitative scores. Data were organised at three hierarchical levels. Journal-level data comprised the publisher, the Journal Impact Factor (JIF), and the cited half-life. Article-

level data comprised the year of publication, the digital object identifier (DOI), and open access status. Measure-level data comprised characteristics of each instrument, including whether it was single-item or multi-item, whether it was newly developed or a modification of an existing measure, and the types of validity and reliability evidence reported. The temporal window of 2002 to 2026 was chosen to span both the pre-reform period and the substantive methodological reform that followed the replication crisis, so that reporting practices before and after this shift could be compared within a single sampling frame.

Procedure

Screening and coder allocation

Although most filtering was embedded within the API call, coders applied two further exclusions at the point of coding. An article was excluded if it did not contain any newly generated or modified psychometric measure, or if it explicitly stated that all relevant measures were not intended for research use. Because coders worked from full-text articles, bibliographic fields such as author names, titles, and abstracts could not be obscured. To limit the bias this introduces, any coder who reported having been involved in the conception, design, analysis, or writing of an assigned article, or who reported any other conflict of interest, was reallocated to a different, randomly selected article. The screening and selection process is summarised in Figure A1 , reported following an adapted PRISMA 2020 flow (Page et al., 2021).

Coding scheme and outcome measures

Each article was coded against a structured scheme in which items and sub-items were scored 1 when the relevant information was reported and 0 when it was not, with no distinction drawn between the main text and supplementary materials. Free-text "other" responses were

scored 1 where one or more entries were provided and 0 where the field was blank. Coded responses were aggregated into five composite scores, whose structure is shown in Figure A2 .

The first composite, scale construction, drew on the items describing how a scale was developed. Multi-response items were scored one point per option selected, and the item recording whether scale items were reproduced was scored on a three-point scale reflecting whether all, example, or no items were provided; this composite ranged from 0 to 14. The second, participant description, aggregated information about the development sample and missing data, treating sample location, administration, and mode as separate sub-items, and ranged from 0 to 8. The third and fourth composites captured validity and reliability: validity ranged from 0 to 2, reflecting whether convergent and discriminant evidence were reported, and reliability from 0 to 4, reflecting the number of distinct reliability indices reported. The fifth, dimensionality, was scored separately for exploratory factor analysis or principal components analysis (0 to 10) and for confirmatory factor analysis (0 to 6), so that articles reporting either or both approaches received the appropriate score.

The primary outcome was the proportion of plausibly reportable information reported, formed by summing the composite scores and dividing by the maximum attainable for that article, which depended on the analytic approaches it adopted. Expressing the outcome as a proportion ensured that articles were evaluated only against the information it was reasonable to expect them to provide.

Intercoder reliability

Reliability of coding was assessed at three points. Before data collection, all coders independently coded a common set of 10 calibration articles, and reliability was reassessed at the midpoint and end of collection using three pre-screened articles each time. Two indices were computed: Krippendorff's alpha and simple percent agreement. Reliability met the preregistered thresholds at every assessment point. Averaged across the three points, percent agreement was 89.75% (range 84.43-94.47) and Krippendorff's alpha was .82 (range .73-.90), both comfortably above the acceptable minima of 75% and 0.67 (Krippendorff, 2019).

During calibration, a small number of coders were directed back to the guidance documents after initial confusion about which measures to code. In these cases they had misunderstood the scope of the task: already-validated measures used without modification were not to be coded, nor were basic demographic questions such as a participant's age or education level. Once clarified, coding continued in line with the agreed scheme.

Analytic plan

The study was designed to answer three research questions. The primary question was whether the overall quality of psychometric reporting for newly developed and modified measures improved over time between 2002 and 2026. The second was whether the type of validity evidence reported, such as dimensionality, internal consistency, convergent validity, discriminant validity, and relations with other variables, has changed over time. The third was whether specific item development procedures, such as expert review, cognitive interviewing, and piloting, have changed over time. The present study’s design and planned analysis was pre-registered in Open Science framework (https://osf.io/q9xz6/overview?view_only=3b2ae45cb6d24f60a47ac2a244d10e00)

Research Question 1

The primary outcome was the proportion of applicable coding-scheme items reported for a given measure, a continuous quantity bounded between 0 and 1. Because a single article can report more than one newly developed or modified measure, an article may contribute several such proportions, which are therefore not independent of one another. To accommodate this structure, the outcome was modelled using a multilevel Bayesian beta regression (Ferrari & Cribari-Neto, 2004; Smithson & Verkuilen, 2006), which is appropriate for continuous proportional data and thus avoids the distortions that arise when bounded outcomes are treated as though they were unbounded and normally distributed, and, through a random intercept for article it accounts for the nesting of measures within articles. To capture change over time while allowing the trajectory to differ before and after the open science reforms that consolidated around 2014 (Open Science Collaboration, 2015) , the fixed part of the linear predictor took a piecewise form with a breakpoint at that year:

$$g(\mu_{ij}) = \ln\left(\frac{\mu_{ij}}{1-\mu_{ij}}\right) = \eta_{ij}$$

$$\eta_{ij} = [\beta_0 + \beta_1 \text{time}_i + \beta_2(\text{time}_i - 12) \times \text{era}_i] + [\mu_{0j} + \varepsilon_{ij}]$$

Here, the subscripts i and j denote an individual measure and the article in which it appears, so that measures (level one) are nested within articles (level two). Time was coded so that 2002 was year zero and each subsequent value represented the number of years elapsed since 2002, and era was a dummy variable coded 0 for the pre-2014 period and 1 for the post-2014 period. Within the fixed part of the model, β_0 is the mean reporting proportion in 2002 in log-odds units, β_1 is the slope of reporting quality across the pre-2014 period, $\beta_1 + \beta_2$ is the slope across the post-2014 period, and β_2 is therefore the difference in slope after 2014 relative to before it, with the two segments meeting at the 2014 breakpoint so that the trend remains

continuous. The random part comprises two terms: μ_{0j} is a by-article random intercept, capturing the degree to which the baseline reporting proportion of article j departs from the overall average and thereby accounting for the non-independence introduced when an article contributes several measures, and ε_{ij} is the level-one residual for measure i within article j . The hypotheses for this question were directional: reporting quality was expected to be broadly flat before 2014, so β_1 was predicted to lie close to zero, and to improve thereafter, so β_2 was predicted to be positive.

Priors were specified on the log-odds scale on which the coefficients are expressed. Considering the plausible range of annual change in a proportion bounded on the unit interval, together with the expectation that reporting should not decline, the slope coefficients were assigned a $N(0.5,0.25)$ prior, which concentrates prior mass over the plausible positive range while permitting modest decreases. The intercept, representing the expected reporting proportion in 2002, was assigned a $N(0,3)$ prior, which comfortably spans base rates from roughly 0.1 to 0.9 in log-odds units, and default weakly informative priors were retained for the dispersion parameter. Because the model includes a random intercept for article, the between-article variance of these intercepts($\sigma^2_{\mu_0}$) must also be estimated as we held no substantive hypothesis or prior information about the magnitude of between-article variability, this parameter was assigned a noninformative prior restricted to positive values.

Research Questions 2 and 3

The individual categories of validity evidence and item development procedures named in Research Questions 2 and 3 were, in most cases, reported too rarely to support separate temporal modelling. These questions were therefore addressed at the level of the coding scheme's sub-dimensions. The piecewise beta-regression model specified for the primary analysis was

refitted separately to each of four sub-dimensions, namely descriptive information about scale construction, participant descriptions, validity and reliability combined, and dimensionality, with articles excluded from the dimensionality model where the relevant factor-analytic items were not reported. Because no directional predictions were made for these sub-dimensions, their slope coefficients were assigned weaker N(0,1) priors.

All models were fitted in R version 4.5.2 (R Core Team, 2025) within a Bayesian framework using the brms interface to Stan (Bürkner, 2017). Posterior distributions were summarised using point estimates and 95% credible intervals, and convergence was assessed through the potential scale reduction factor and effective sample size for all parameters.

Ethics

The study analysed published, publicly accessible bibliographic and article data and did not require review by a research ethics committee. All source articles were obtained through legitimate institutional and open-access routes, and no personal or identifying data were collected. In keeping with open and reproducible research practice, the study's design, hypotheses, and analysis plan were pre-registered, and the coding materials and analysis code will be made openly available. The plan was specified before the analysis but, owing to a technical error, was not successfully submitted until afterwards, making the registration retrospective in submission; the analysis follows that plan without alteration.

Results

The final analytic sample comprised 1,142 psychometric measures nested within 530 articles. Every coded article matched its journal and publication metadata, and none fell outside the 2002 to 2026 study window. Across the full sample, the overall proportion of reporting criteria met was low. On average, measures met 41% of the coded criteria (SD = 0.11), with

scores ranging from 0.03 to 0.75. No measure in the sample met more than three quarters of the applicable reporting criteria. Reporting differed substantially across the four sub-dimensions. Sample description was the most completely reported, with measures meeting on average 73% of its criteria (SD = 0.20). Dimensionality was reported at an intermediate level (M = 0.46, SD = 0.22), computed only for the 248 measures that reported a factor analytic procedure, while scale construction was reported at a similar level to the overall score (M = 0.37, SD = 0.15). Validity and reliability evidence was the least reported by a considerable margin, with measures meeting on average only 13% of its criteria (SD = 0.14). Full descriptive statistics for the overall score and the four sub-dimensions are presented in Table A2 . The clustering of measures within articles is illustrated in Figure A3, which shows the number of measures contributed by each article. While most articles reported a single measure, a substantial number reported two or more, and a small number reported many. This distribution confirmed that treating measures as independent observations would misrepresent the structure of the data, motivating the inclusion of a random intercept for article in all models.

Convergence and fit were then assessed for the primary model. All coefficients returned R-hat values at or below 1.003, indicating that the four chains had converged to a common posterior distribution, and bulk effective sample sizes exceeded 2,100 for every parameter (see Table A5), well above the threshold recommended for stable estimation of posterior summaries (Vehtari et al., 2021). Visual inspection of the trace plots confirmed this, showing four dense, overlapping, and stationary chains for each parameter with no drift or divergence. A posterior predictive check (as seen in Figure A6 and Figure A7) indicated close correspondence between the observed distribution of reporting proportions and the distributions simulated from the fitted model, suggesting that the beta likelihood captured the shape of the outcome adequately across

the range of the scale. The four secondary sub-dimension models were subjected to the same checks and likewise converged satisfactorily, with R-hat values at or below 1.008 and effective sample sizes exceeding 1,000 for every coefficient.

Parameters (as seen in Table A3)were summarised using the median of the posterior distribution, the 95% highest density credible interval, the probability of direction as an index of the existence of an effect, and the proportion of the posterior falling within a region of practical equivalence to zero (Makowski et al., 2019) .

Primary Analysis

The primary research question asked whether the overall quality of psychometric reporting changed across the study period. Neither temporal parameter reached credibility. The pre-2014 slope was negligible (β = 0.005, 95% CrI [−0.010, 0.019]), with a probability of direction of 72.62% and the entire credible interval falling within the region of practical equivalence, indicating no credible trend before 2014. The change in slope from 2014 onward, which constituted the direct test of whether reporting improved following the reporting standards revision and the reform movement, was likewise negligible (β= −0.001, 95% CrI [−0.024, 0.021]). Its probability of direction was 53.54%, barely distinguishable from chance, and its credible interval also fell entirely within the region of practical equivalence. The directional hypothesis that the post-2014 slope change exceeded zero was not supported (evidence ratio = 0.87), and the hypothesis that the post-2014 slope itself was positive was similarly unsupported. Because both credible intervals crossed zero and lay wholly within the region of practical equivalence, the finding is best interpreted not as an absence of evidence for change, but as positive evidence for the absence of change (Kruschke, 2018). Full parameter estimates are reported in Table A3 , and the modelled trend is shown in Figure A4 . The variance

decomposition reinforced this. The temporal predictors alone accounted for almost none of the variance in reporting quality (marginal R^2 = 0.007, 95% CrI [0.000, 0.021]), whereas the full model including the by-article random intercept accounted for the large majority (conditional R^2 = 0.798, 95% CrI [0.779, 0.815]), indicating that variation in reporting quality was attributable overwhelmingly to differences between articles rather than to publication year.

Secondary Analysis

The first secondary question asked whether the reporting of specific types of validity evidence changed over time. Neither the validity and reliability sub-dimension nor the dimensionality sub-dimension showed a credible temporal effect. Validity and reliability reporting was flat before 2014 (β = −0.011, 95% CrI [−0.035, 0.013]). Its post-2014 change in slope carried the highest probability of direction of any coefficient in the analysis (β = 0.030, 95% CrI [−0.005, 0.066], pd = 95.13%) and leaned positive, but it did not reach the threshold for a credible effect and its interval remained within the region of practical equivalence. This may be read as a faint and non-credible suggestion that reporting of validity and reliability evidence rose modestly after 2014, from the lowest baseline of any sub-dimension, but no confident conclusion is warranted. Dimensionality reporting, estimated on the subset of measures reporting a factor analysis, was also flat in both eras (pre-2014 β = −0.020, 95% CrI [−0.079, 0.040]; post-2014 β = 0.026, 95% CrI [−0.063, 0.116]).

The second secondary question asked whether the reporting of scale construction and development changed over time. Scale construction showed no credible change in either era (pre-2014 β = 0.008, 95% CrI [−0.015, 0.029]; post-2014 β = −0.014, 95% CrI [−0.046, 0.020]), and neither did sample description (pre-2014 β = −0.008, 95% CrI [−0.038, 0.021]; post-2014 β = 0.022, 95% CrI [−0.023, 0.068]). In every case the credible interval crossed zero and fell within

the region of practical equivalence. Parameter estimates for all four sub-dimension models are reported in Table A4 , and their trajectories across the study period are shown in Figure A5 .

Because the individual development procedures named in Research Question 3 were captured within the scale-construction composite, their reporting was additionally examined descriptively. All five procedures were reported for only a small minority of measures in both eras, and none showed a systematic change across the 2014 breakpoint. Review of existing measures was the most frequently reported (20.6% of measures pre-2014, 16.7% post-2014), followed by piloting (9.1%, 5.6%) and expert review (5.4%, 7.8%), while focus groups (3.7%, 2.5%) and cognitive interviews (0.3%, 1.0%) were reported for very few measures throughout. Full counts and proportions are given in Table A6.

Discussion

The present study addressed whether the quality of psychometric reporting for newly developed and modified measures in psychology has changed over time. Across a multi-subfield sample of 1,142 measures nested within 530 articles, modelled with a Bayesian multilevel beta regression treating publication year as a piecewise function with a breakpoint fixed at 2014, the result was consistent: reporting quality did not change. The overall proportion of criteria met remained stable throughout, with no credible trend before 2014 and no credible shift at the breakpoint marking the standards revision and reform consolidation. This held across all four sub-dimensions, none of which showed a credible temporal effect in either era. The one coefficient approaching conventional thresholds, the post-2014 change in validity and reliability reporting, leaned positive but did not reach credibility and stayed within the region of practical equivalence to zero, at most a faint, non-credible suggestion of increase from the lowest baseline of any sub-dimension.

These findings supported neither the primary hypothesis of improved overall reporting quality nor either secondary hypothesis concerning validity evidence or development procedures. Because the analysis was Bayesian, this flatness can be asserted rather than merely inferred from a failure to reject: the probability of direction for the primary post-2014 slope was close to chance, and its credible interval fell wholly within the region of practical equivalence, together supporting a positive conclusion of no change rather than the weaker claim that change went undetected (Kruschke, 2018; Makowski et al., 2019). This matters because a finding of stability is only compelling if the analysis can distinguish genuine absence from an underpowered null, which the precision of the present estimates confirms.

The beta likelihood, used in preference to a Gaussian model, strengthens this inference further. Since the outcome is a proportion bounded between zero and one, a Gaussian model would assign probability to impossible values beyond the scale, distorting both the estimated trend and its uncertainty, most severely for sub-dimensions whose scores clustered near a boundary (Bürkner, 2017). This mattered most for validity and reliability, where many measures scored at the floor and a boundary-insensitive model would have misrepresented reporting in precisely the domain where it was weakest.

The most revealing feature of the analysis, however, was the partition of variance between the temporal predictors and the article-level random intercept. Publication year explained a negligible fraction of the variance in reporting quality, whereas differences between articles explained the large majority. The dominant determinant of how well a measure is reported is therefore not the era of publication but the article in which it appears, and by extension the practices of its authors and the standards of its outlet.

Situating the finding in the measurement literature

Because no prior study has modelled psychometric reporting quality across so broad a sample or so long a window, direct comparison is limited. The most closely comparable work is Lawson et al. (2026), who tracked measurement-reporting practices across a decade in a single social and personality journal and found that most did not improve, the salient exception being the sharing of study materials. The present findings are consistent with theirs and extend them substantially, across five subfields rather than one, two decades rather than one, and the more demanding categories of psychometric evidence the earlier study identified as consequential but did not code. That two studies using different samples and operationalisations converge on the same conclusion lends it a robustness neither secures alone, and suggests that the stagnation of measurement reporting is a general property of the field rather than a peculiarity of a particular journal or coding scheme.

The findings sit less comfortably alongside the broader narrative of a credibility revolution (Korbmacher et al., 2023), under which the field's methodological culture is held to have improved markedly since the replication crisis. Whatever the reach of that improvement in analytic transparency and data sharing, it did not extend to the reporting of the instruments on which quantitative claims depend. This is intelligible in light of how the reform movement defined its targets. The questionable research practices that drove reform were predominantly analytic, concerning the flexibility available in the treatment of data once collected (Nagy et al., 2025; Simmons et al., 2011). Questionable measurement practices, by contrast, concern the flexibility available in the construction and selection of the instruments that generate the data, and although Flake and Fried (2020) argued that these are at least as consequential, they have received less attention from reformers. The blind spot is therefore structural: reform concentrated

on what researchers do with data and largely bypassed whether the data mean what they are taken to mean.

Why has sustained awareness produced no change?

The more searching question these results pose is why. Deficiencies in psychological measurement have been documented and named repeatedly, from the reviews of the 1990s to the influential critiques of the past decade (Flake et al., 2017; Flake & Fried, 2020; Hussey & Hughes, 2020) , and the relevant standards and reporting guidance have long been available (American Educational Research Association, et al., 2014; Schubert et al., 2025) . That such sustained awareness has not been accompanied by any measurable improvement is the central question these findings raise, and several explanations are worth considering.

The first is that the reforms that succeeded, foremost the sharing of study materials, were inexpensive, visible, and increasingly mandated, whereas the thorough reporting of validity evidence is effortful, competes for limited space, and is seldom a condition of acceptance. Lawson et al. (2026) observed that the practices which improved were the least demanding, and that the more substantial forms of evidence remained absent where they were most needed. Where reform has depended on the voluntary modification of individual behaviour, its uptake has been more uneven than where change has been driven by structural and community-level mechanisms, which have been argued to underpin the more durable improvements of the reform era (Korbmacher et al., 2023).

The second one is that measures may frequently be used without rigorous validation, so that improved reporting cannot follow because there is nothing to disclose. This draws indirect support from evidence that shortcomings persist even when researchers are observed working rather than writing. Sladekova et al. (2026) examining how 188 researchers conducted two

analytic tasks, found that checks for basic modelling assumptions were routinely omitted or performed in ways incapable of detecting the problems present, with only a modest advantage among those who taught research methods. Sparse reporting is often a faithful record of sparse underlying practice and remedying it would require changing what researchers do rather than only what they declare.

A third, deeper explanation situates measurement within psychology's broader theoretical difficulties. Eronen and Bringmann (2021) argued that the field's recurrent crises stem partly from poorly specified and validated constructs, since vaguely defined constructs cannot be measured well no matter how carefully reporting is done. On this view, improved reporting is necessary but not sufficient: it cannot compensate for constructs whose theoretical specification is too weak to support valid measurement in the first place (Flores-Kanter, 2026). The stability observed here may therefore reflect not just institutional inertia, but a problem reporting reform alone cannot structurally solve.

Recommendations for practice

The findings carry implications for editorial practice and the culture of measurement in psychology. If two decades of awareness and reform have failed to move measurement reporting, the voluntary and educational mechanisms relied upon to date are insufficient, and structural approaches are indicated. The single reform that succeeded in the comparable literature did so because it became an expectation enforced at submission (Lawson et al., 2026). Echoing this, the present findings recommend that journals adopt a mandatory measurement-reporting checklist, verified during peer review, requiring authors either to state the validity and reliability evidence for each measure or explicitly to declare its absence. The declaration of absence itself is valuable, converting a silent omission into a visible one that readers and reviewers can weigh.

Because the variance decomposition located the determinants of reporting quality at the level of articles and outlets rather than the field as a whole, editorial enforcement is well targeted: a problem residing in the practices of particular authors and journals is more amenable to gatekeeping than one diffusing across the discipline.

A second recommendation concerns the point of intervention. If sparse reporting reflects sparse practice, reporting mandates alone will be insufficient, since they cannot summon evidence that was never generated. Intervention is therefore warranted earlier in the research lifecycle, at the stage of measure selection and construction, alongside infrastructure that makes rigorous validation less costly. Validated measure repositories, registered measurement-development reports analogous to registered reports, and normative expectations that new measures undergo and document formal validation before use would each address the problem at its source (Clark & Watson, 2019). Given the proliferation of ad hoc and modified measures across the journal categories in the present sample, such infrastructure is likely to become more necessary over time (Anvari et al., 2025).

Limitations and future directions

A key limitation concerns interpreting reporting quality as a proportion of a fixed, preregistered criterion set. Although the maxima were specified a priori to represent what could feasibly be reported, the results are conditional on that set, and the upper region of the scale may rarely have been reached in practice, since no measure exceeded three quarters of the criteria. This is compatible both with genuinely incomplete reporting and with a ceiling that was seldom attainable. Future research could validate the criterion set against exemplar measures known to be well reported, establishing whether the ceiling is reachable and calibrating the scheme accordingly.

A related limitation arises from the conditional structure of the dimensionality items, several of which were displayed only when a factor analysis extracting more than one factor was reported. Measures reporting a single-factor solution could not satisfy these criteria, and treating the unshown items as unmet may have understated their reporting quality. Future coding schemes would benefit from scoring conditional items against a denominator restricted to those applicable to each measure.

A third and more fundamental limitation concerns the distinction between the transparency of reporting and the substantive quality of the measurement reported. The outcome quantifies whether information was reported, not whether the evidence, where present, was adequate: a measure could report a factor analysis, a reliability coefficient, and validity evidence, and thereby score highly, while those analyses were poorly conducted. The study therefore speaks to reporting transparency rather than measurement quality, and the two must not be conflated. Future research could assess the substantive adequacy of reported evidence rather than its mere presence, separating the question of whether measurement itself has improved from whether its reporting has.

A final limitation is inferential. The piecewise model tests for a change in the rate of improvement at 2014, a specific assumption about the shape of any temporal effect. A reform influence that accumulated gradually, operated with a delay, or altered the variance rather than the mean of reporting might not appear as the slope change the model was designed to detect. Two features of the results ease this concern: no trend emerged in either period, so the stability holds across the whole window rather than only around the breakpoint, and the credible intervals for both slopes were narrow and fell within the region of practical equivalence, limiting any

undetected effect to a negligible size. Future work could extend this with a change-point model that estimates the location of any shift rather than fixing it.

Conclusion

The present study has addressed a significant gap in the literature concerning whether psychometric reporting quality for newly developed and modified measures has changed over time. A Bayesian multilevel beta regression indicated no credible change in reporting quality across the period 2002 to 2026, either overall or within any of its four sub-dimensions, and no detectable improvement at the breakpoint marking the standards revision and the reform period. Reporting was uniformly low, and its baseline differed markedly across sub-dimensions while remaining stable within each over two decades. The variance decomposition further indicated that reporting quality was determined more by the article and its outlet than by the era of publication. These findings have strong implications for editorial and research practice, indicating that the methodological reforms of the past decade, whatever their success elsewhere, did not reach the reporting of measurement, and that the awareness-based mechanisms relied upon to date have proved insufficient. It is our recommendation that structural interventions, in the form of enforced reporting requirements at submission and infrastructure supporting validation earlier in the research process, are now required. Given how fundamental sound measurement is to the validity of psychological findings, it is our hope that the present study will provide the impetus for future research to examine these interventions directly, and to distinguish the reporting of measurement quality from its substance, so that the field's understanding of its most basic practices can at last begin to improve.

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Appendix

Figure A1

Screening and selection of articles, following an adapted PRISMA 2020 flow.

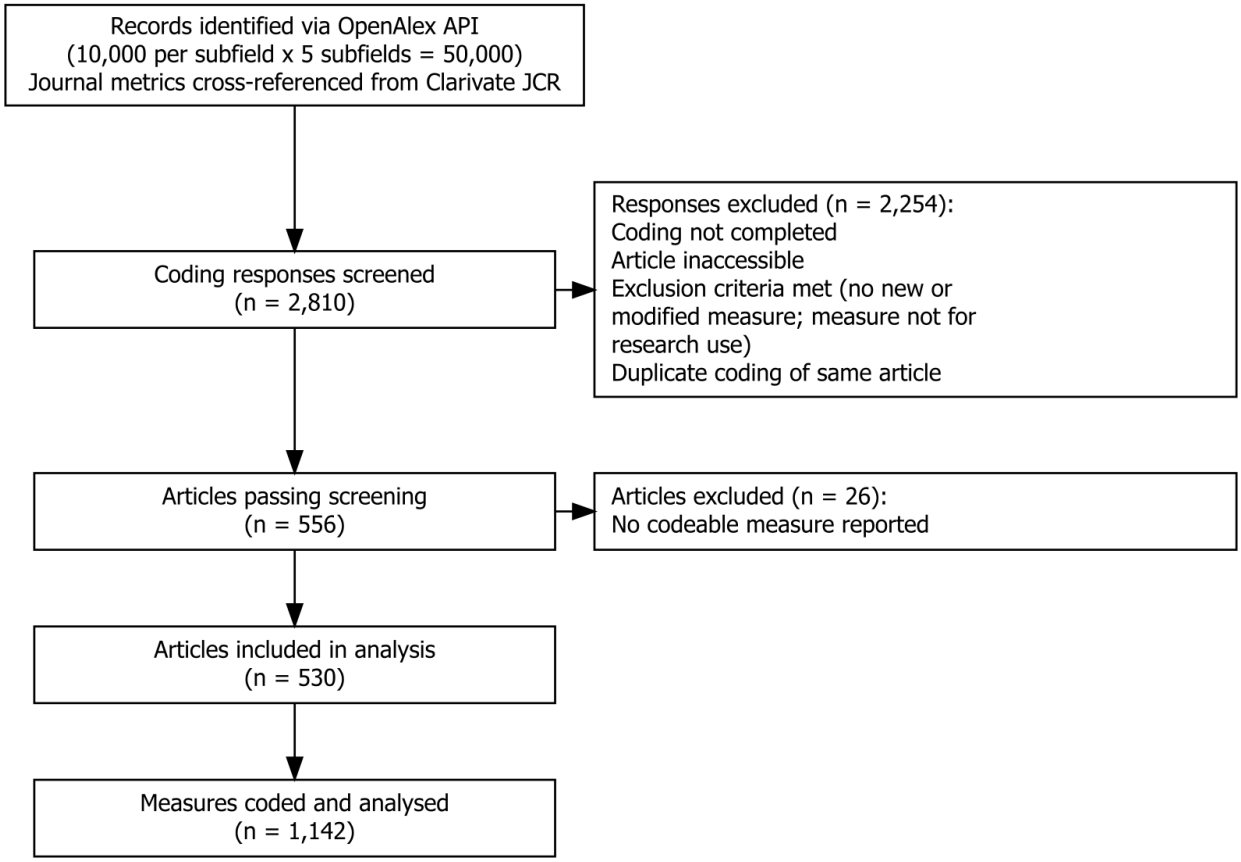


Figure A2

Structure of the coding scheme and its scoring.

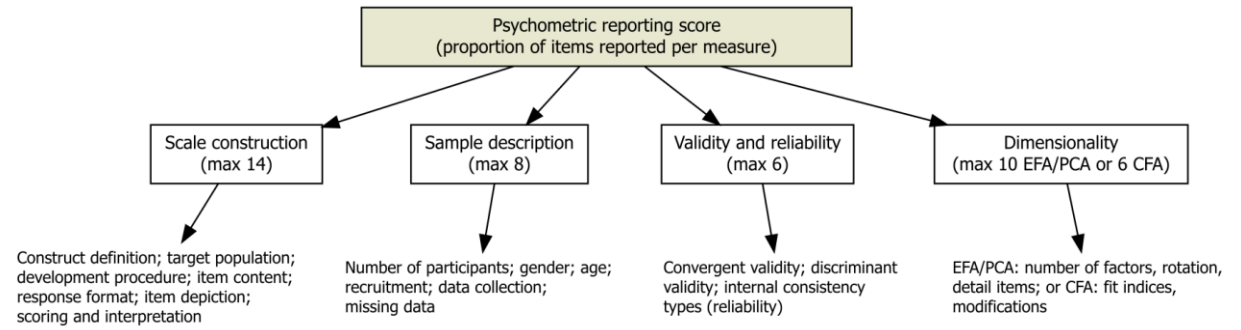


Table A1

Distribution of Analysed Articles Across Journal Citation Reports Categories

category	n_articles
Psychology, Multidisciplinary	64
Education & Educational Research	52
Psychiatry	52
Psychology, Clinical	37
Psychology, Social	35
Psychology, Experimental	32
Psychology, Developmental	29
Psychology	27
Health Care Sciences & Services	18
Psychology, Applied	18
Psychology, Educational	18
Criminology & Penology	17
Nursing	17
Education, Special	11
Hospitality, Leisure, Sport & Tourism	11
Behavioral Sciences	10
Social Work	10
Communication	9
Social Sciences, Interdisciplinary	9
Sport Sciences	8
Education, Scientific Disciplines	5
Gerontology	4
Health Policy & Services	4
Environmental Studies	3

category	n_articles
Ethnic Studies	3
Industrial Relations & Labor	3
Law	3
Political Science	3
Sociology	3
Substance Abuse	3
Ethics	2
Social Issues	2
Transportation	2
Womens Studies	2
Economics	1
Geography	1
Public Administration	1
Social Sciences, Mathematical Methods	1

Table A2

Descriptive statistics for the overall reporting proportion and the four sub-dimensions

Variable	Mean	SD	IQR	Range	n	n_Missing
prop_score	0.41	0.11	0.13	(0.031, 0.750)	1142	0
prop_construction	0.37	0.15	0.18	(0.000, 0.882)	1142	0
prop_participants	0.73	0.20	0.25	(0.125, 1.000)	1142	0
prop_valrel	0.13	0.14	0.14	(0.000, 0.714)	1142	0
prop_dimensionality	0.46	0.22	0.44	(0.000, 0.889)	248	894

Figure A3

Distribution of the number of measures reported per article

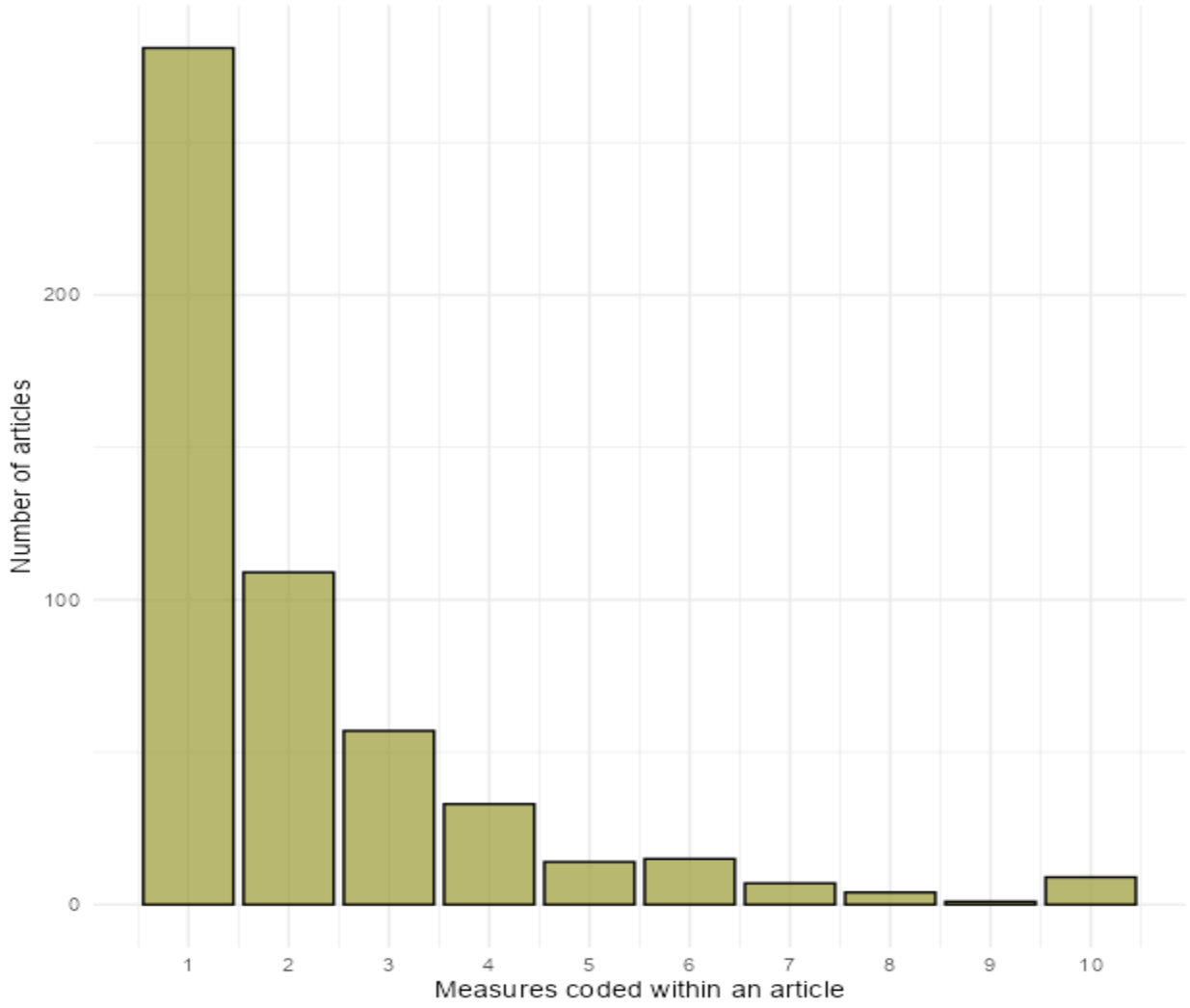


Table A3

Parameter estimates for the primary Bayesian multilevel beta model

Parameter	Effects	Median	CI	CI_low	CI_high	pd	ROPE_Percentage	Rhat	ESS_tail	Group
b_Intercept	fixed	-0.4096282	0.95	-0.5481979	-0.2661113	1.000000	0	1.000753	4801	
b_time	fixed	0.0046942	0.95	-0.0103944	0.0193281	0.726250	1	1.002012	4632	
b_time_post	fixed	-0.0011093	0.95	-0.0244545	0.0214142	0.535375	1	1.002617	4514	
sd_doi__Intercept	random	0.4235972	0.95	0.3948276	0.4556513	1.000000	0	1.000494	6761	doi

Note. Marginal R^2 = .007, 95% CrI [.000, .021]; conditional R^2 = .798, 95% CrI [.779, .815].

Figure A4

Modelled trend in overall reporting quality across 2002 to 2026, with the piecewise fit and 2014 breakpoint

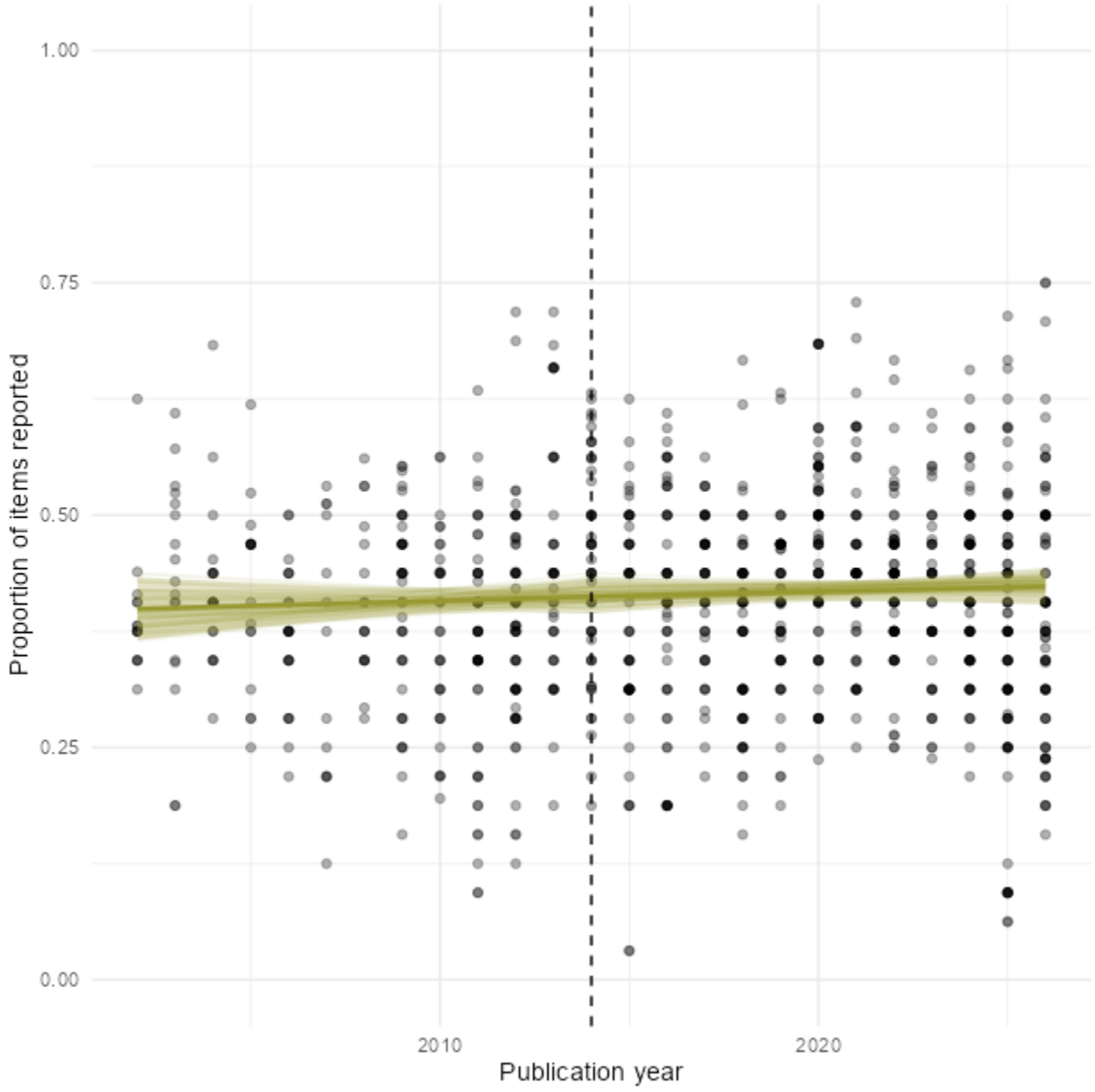


Table A4

Parameter estimates for the four sub-dimension models

outcome	Parameter	Median	CI_low	CI_high	pd	ROPE_Percentage	Rhat	ESS_tail
construction	b_Intercept	-0.518	-0.732	-0.325	1.000	0.000	1.001	7791
	b_time	0.007	-0.015	0.029	0.759	1	1.001	6811
	b_time_post	-0.014	-0.046	0.02	0.792	1	1.000	6529
participants	b_Intercept	0.931	0.654	1.201	1.000	0.000	1.001	3318
	b_time	-0.008	-0.038	0.021	0.701	1	1.002	2900
	b_time_post	0.022	-0.023	0.068	0.828	1	1.008	2683
validity_rel	b_Intercept	-1.432	-1.658	-1.201	1.000	0.000	1.000	5471
	b_time	-0.011	-0.035	0.013	0.812	1	1.000	5184
	b_time_post	0.03	-0.004	0.066	0.951	1	1.001	5142
dimensionality	b_Intercept	0.109	-0.425	0.671	0.647	0.473	1.004	3543
	b_time	-0.020	-0.078	0.04	0.744	1	1.003	3089
	b_time_post	0.026	-0.063	0.116	0.718	1	1.003	2796

Figure A5

Reporting trajectories for the four sub-dimensions across 2002 to 2026

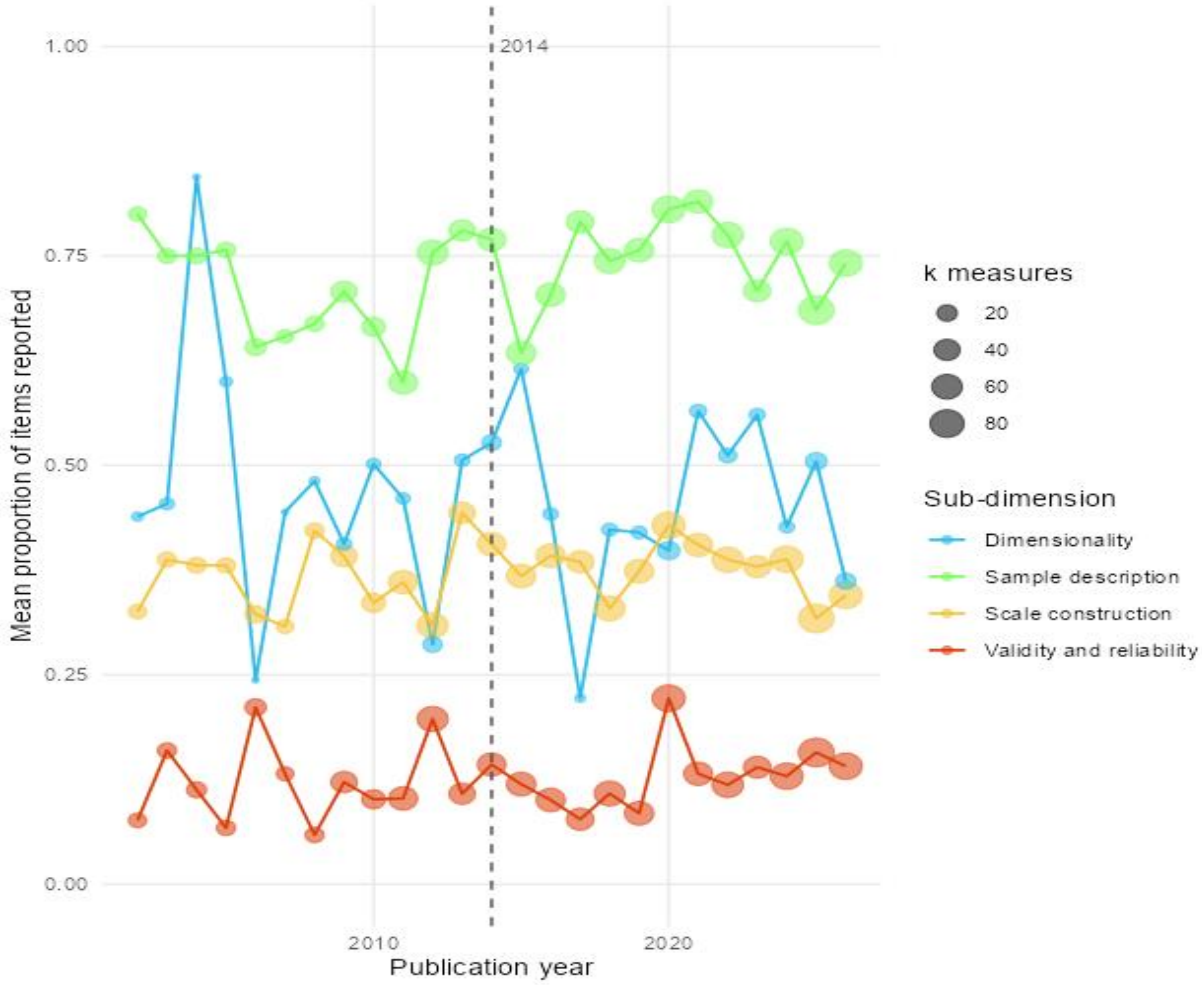


Figure A6

Trace plots and posterior predictive check for the primary model

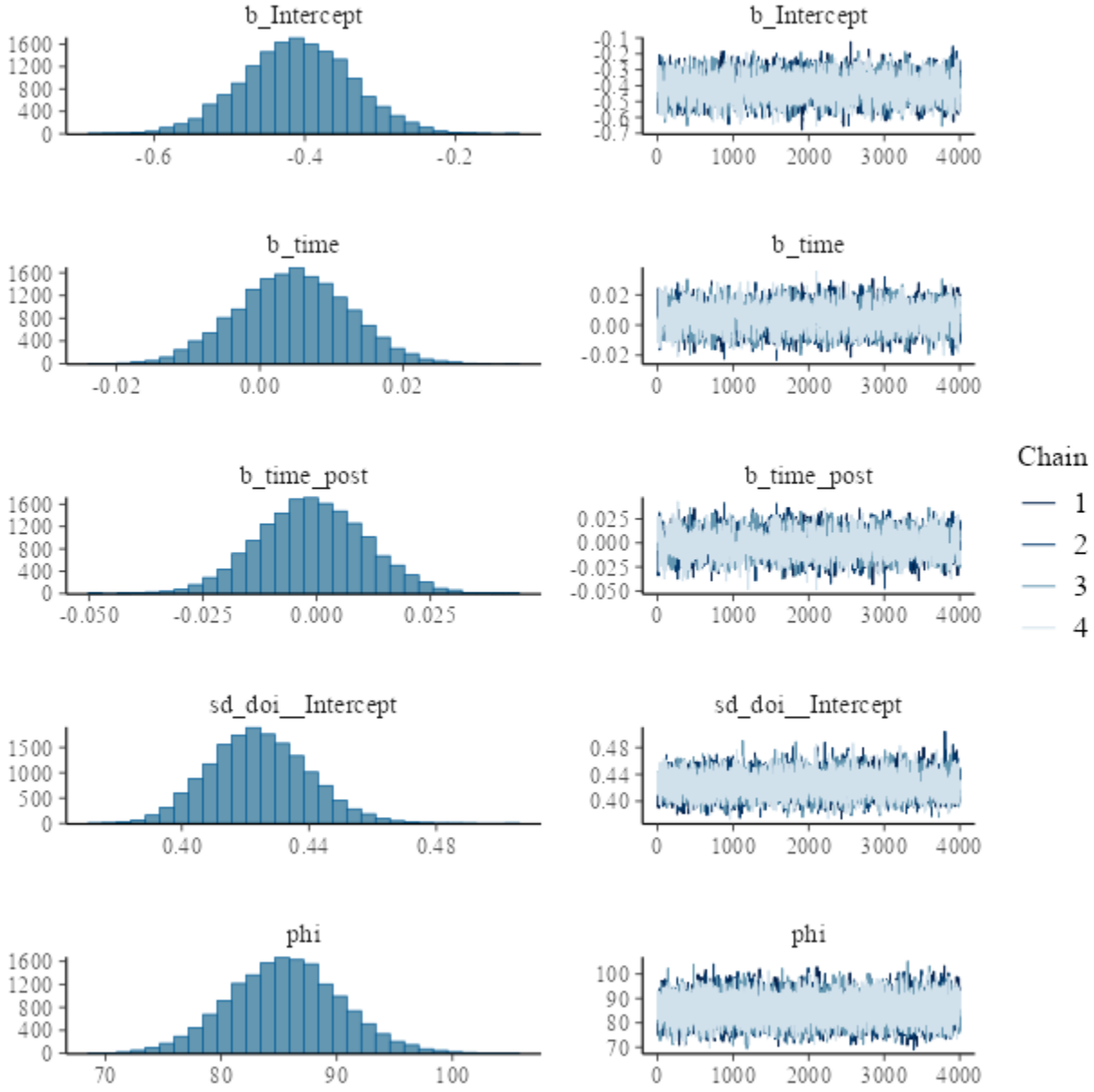


Figure A7

Posterior predictive check for the primary model

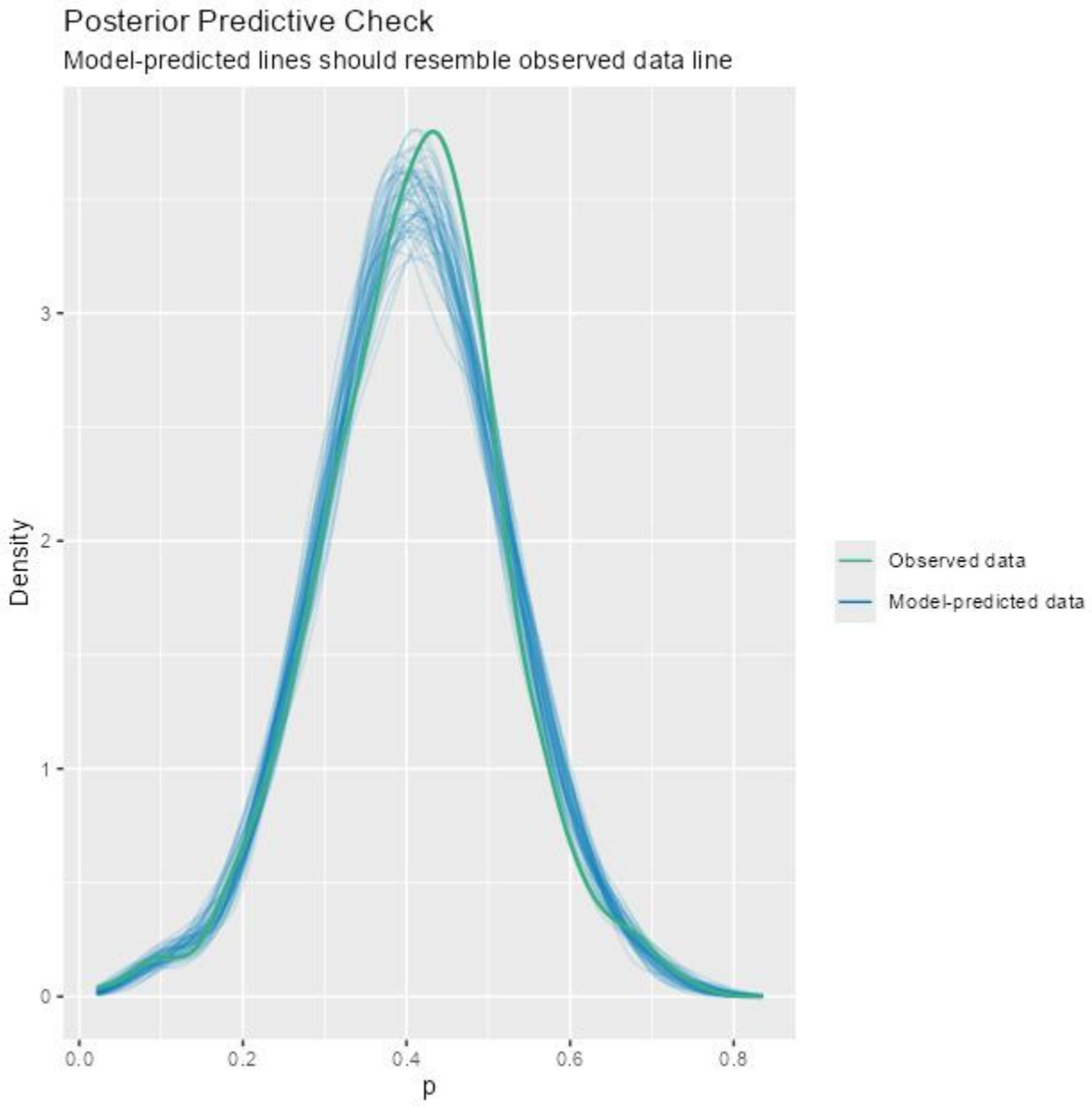


Table A5

Diagnostic Parameters of the Posterior Distribution

Parameter	Rhat	ESS_tail	ESS_bulk	MCSE
b_Intercept	1.000753	4801	2395	0.0016388
b_time	1.002012	4632	2214	0.0001808
b_time_post	1.002617	4514	2161	0.0002897

Table A6

Proportion of Measures Reporting Each Development Procedure by Era

Era	n	Review of literature/existing measures	Focus groups	Cognitive interviews	Expert review	Piloting	Other
Post-2014	792	16.70%	2.5%	1.0%	7.8%	5.6%	7.2%
Pre-2014	350	20.60%	3.7%	0.3%	5.4%	9.1%	6.0%